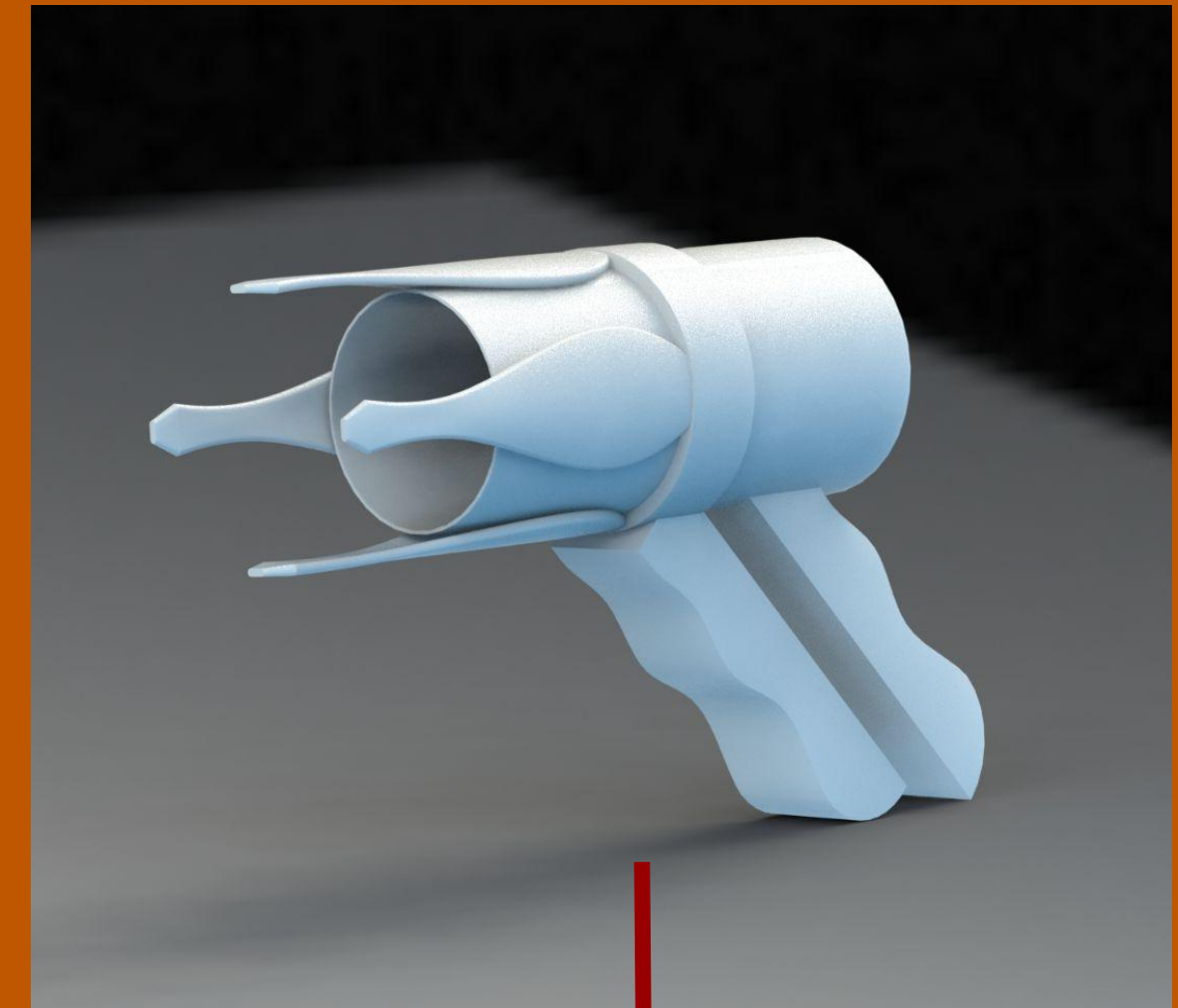


Medical Mavericks: Patient Centered Speculum

Giving control and autonomy of cervical healthcare back to the patient to decrease discomfort and anxiety for a better patient examination experience.

Akash Jape, 2027 Mechanical Engineering
Adhishree Luintel, 2027 Biomedical Engineering
Vy Nguyen, 2024 Biomedical Engineering
Zeynep Yakay, 2026 Biomedical Engineering
Marisa Ballard, 2024 Biomedical Engineering

Version V1



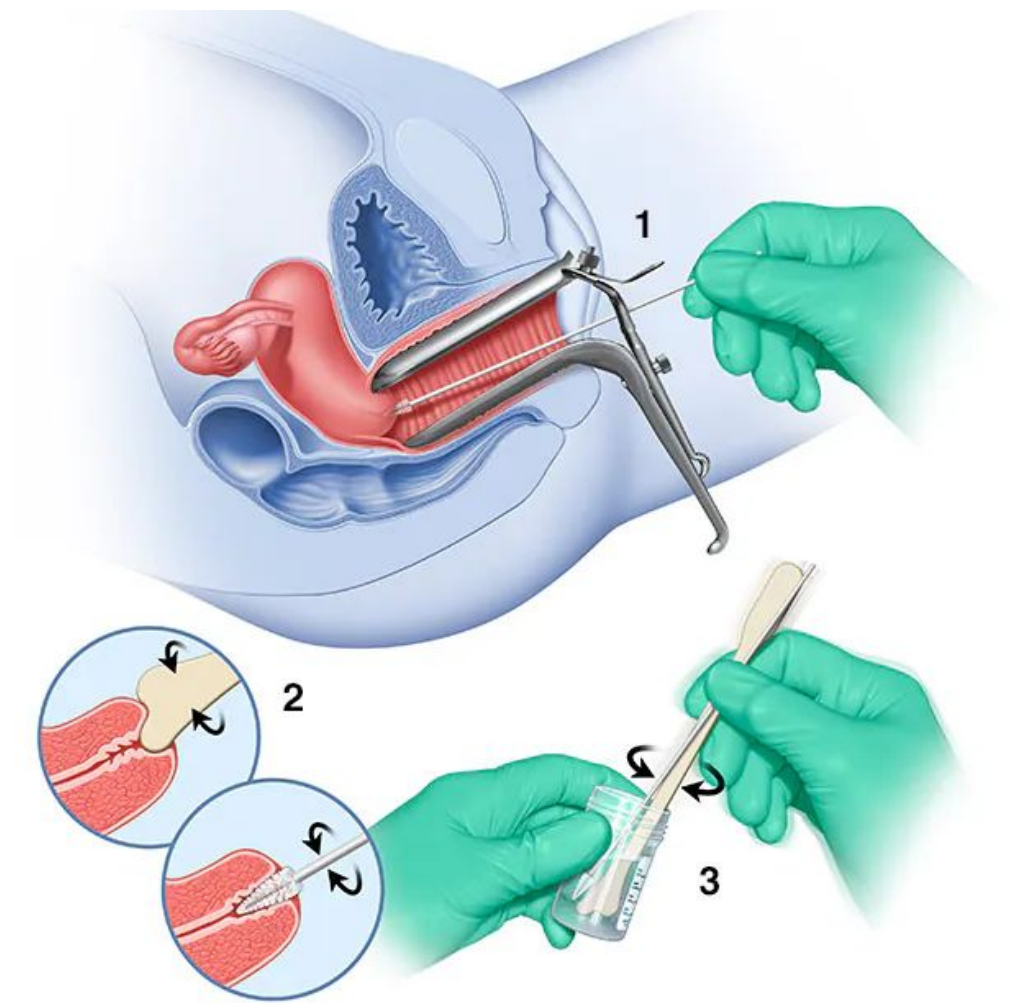
Version V2



Background: Pap Smear

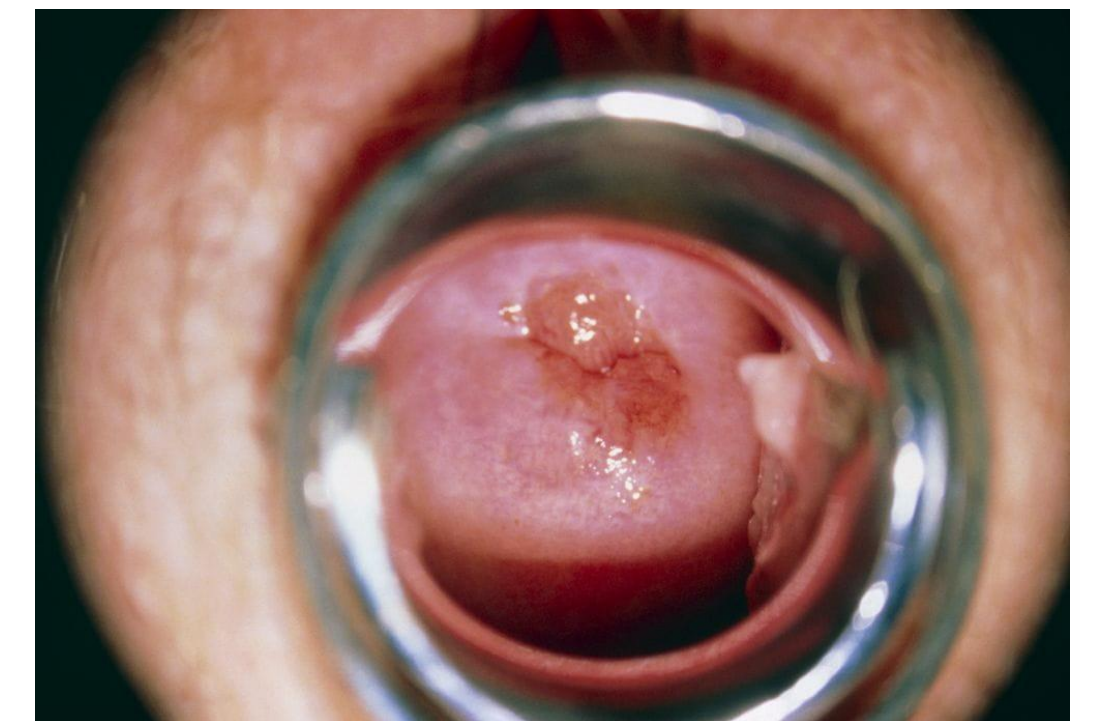
- Pap smears are a common way of screening for cervical cancer, the fourth most common malignancy in women worldwide¹
- Pap smears are recommend to be complete **every three years** from **ages 21- 65**¹
 - *More than **96.5 million** women are between the ages of 21-65 in the US*²
- Screening programs have been shown to **reduce** the incidence of cervical cancer in developing countries

Insertion of Speculum / Swab



<https://www.mayoclinic.org/tests-procedures/pap-smear/multimedia/pap-test/img-20007025>

Image of Cervix with Cancer

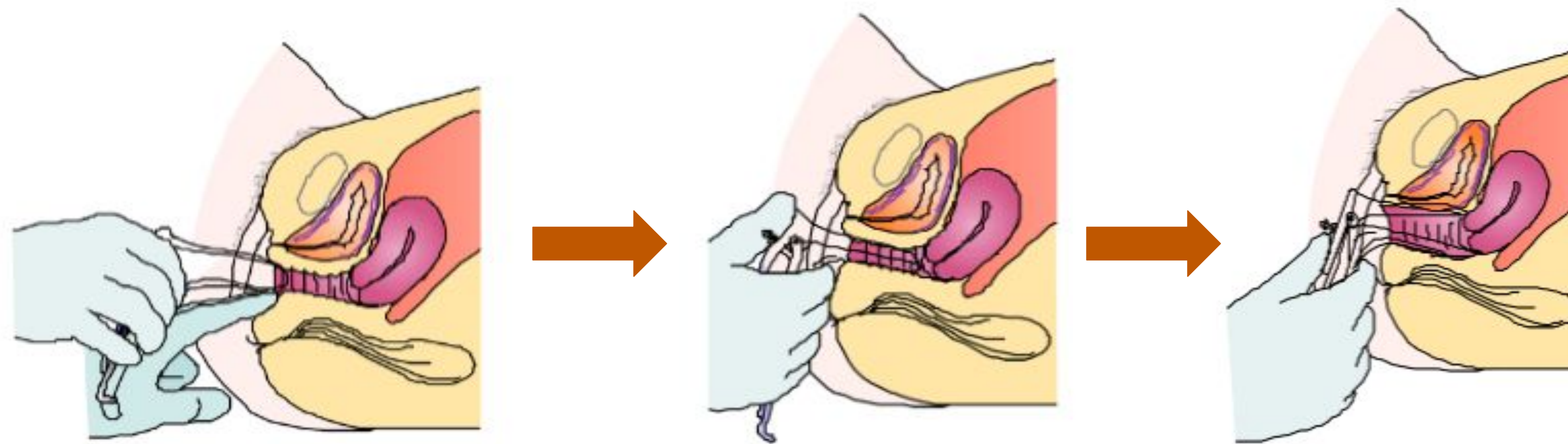


<https://www.merckmanuals.com/professional/gynecology-and-obstetrics/gynecologic-tumors/cervical-cancer>

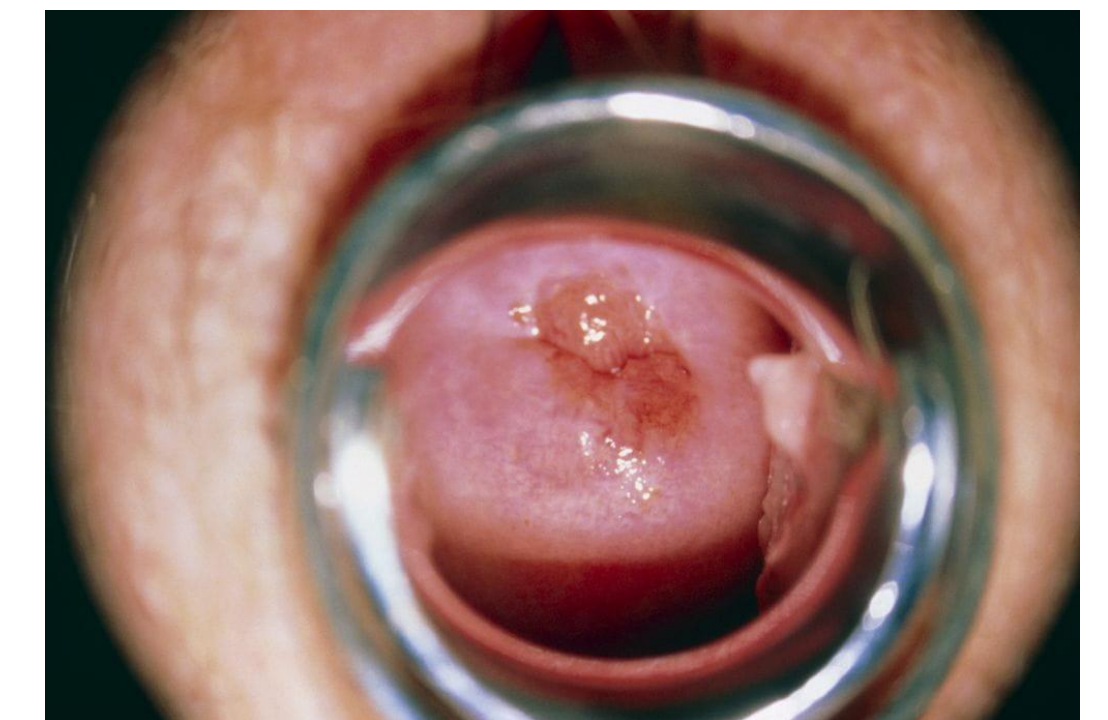
1) <https://pubmed.ncbi.nlm.nih.gov/30064936/>

2. US Census Bureau

- ## Insertion Procedure of the Speculum



View of the cervix using a speculum



1. <https://pubmed.ncbi.nlm.nih.gov/30064936/> 2. U.S. Department of Health and Human Services

Problematic Experiences

A) The procedure!

B) Fear intimidation of cervical exams¹

- 21%-64% of women felt anxiety²
- 52% of women felt embarrassment²
- 22%-68% of women felt pain²

C) The provider is not aware of patient's discomfort³

- Patient's existing health conditions
- Provider's expertises



Market Analysis

Disposable
Speculum
1\$



Pederson
Speculum
24\$



Other Speculums
+100\$

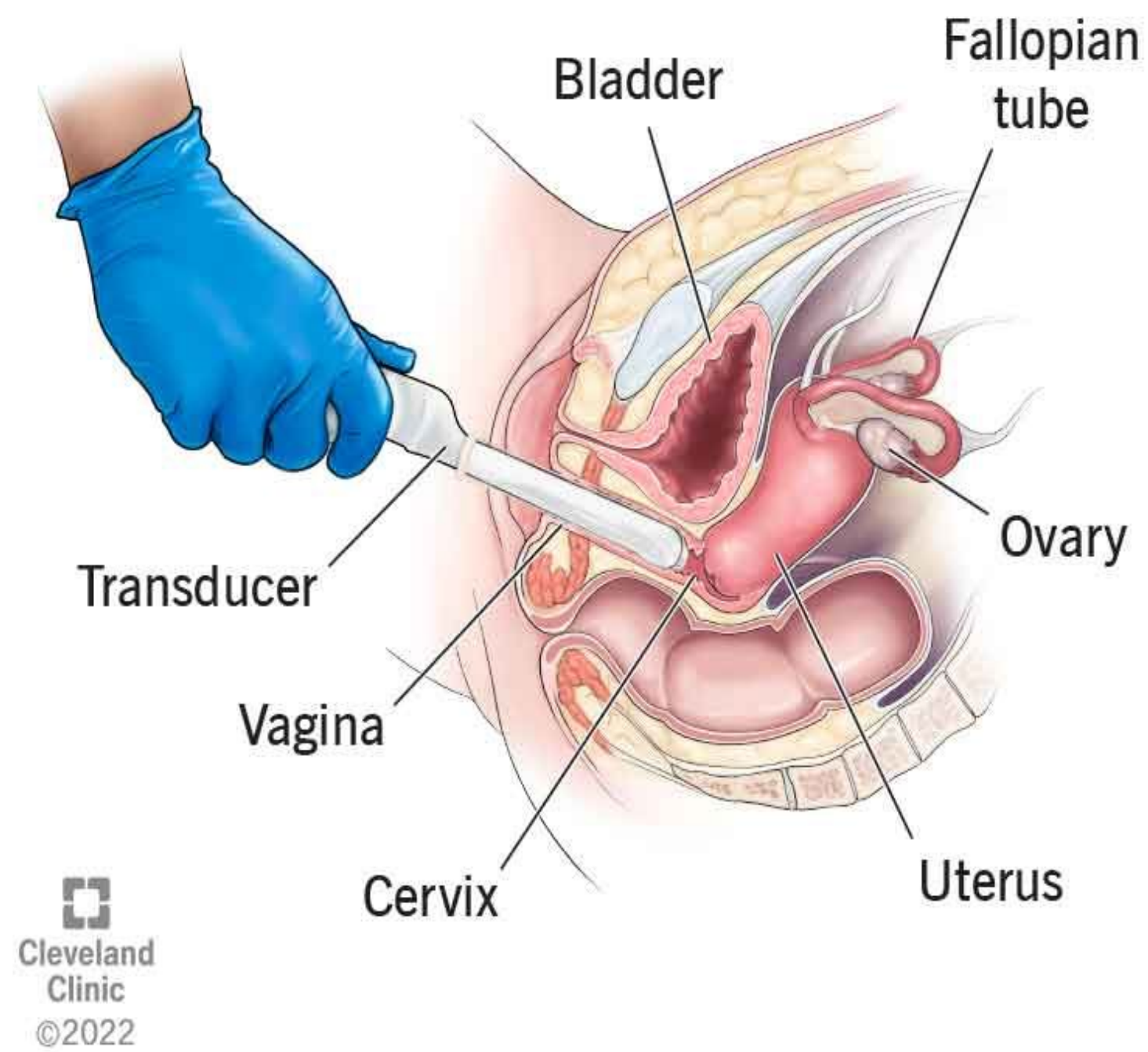


Market Analysis



Patient Controlled Procedure

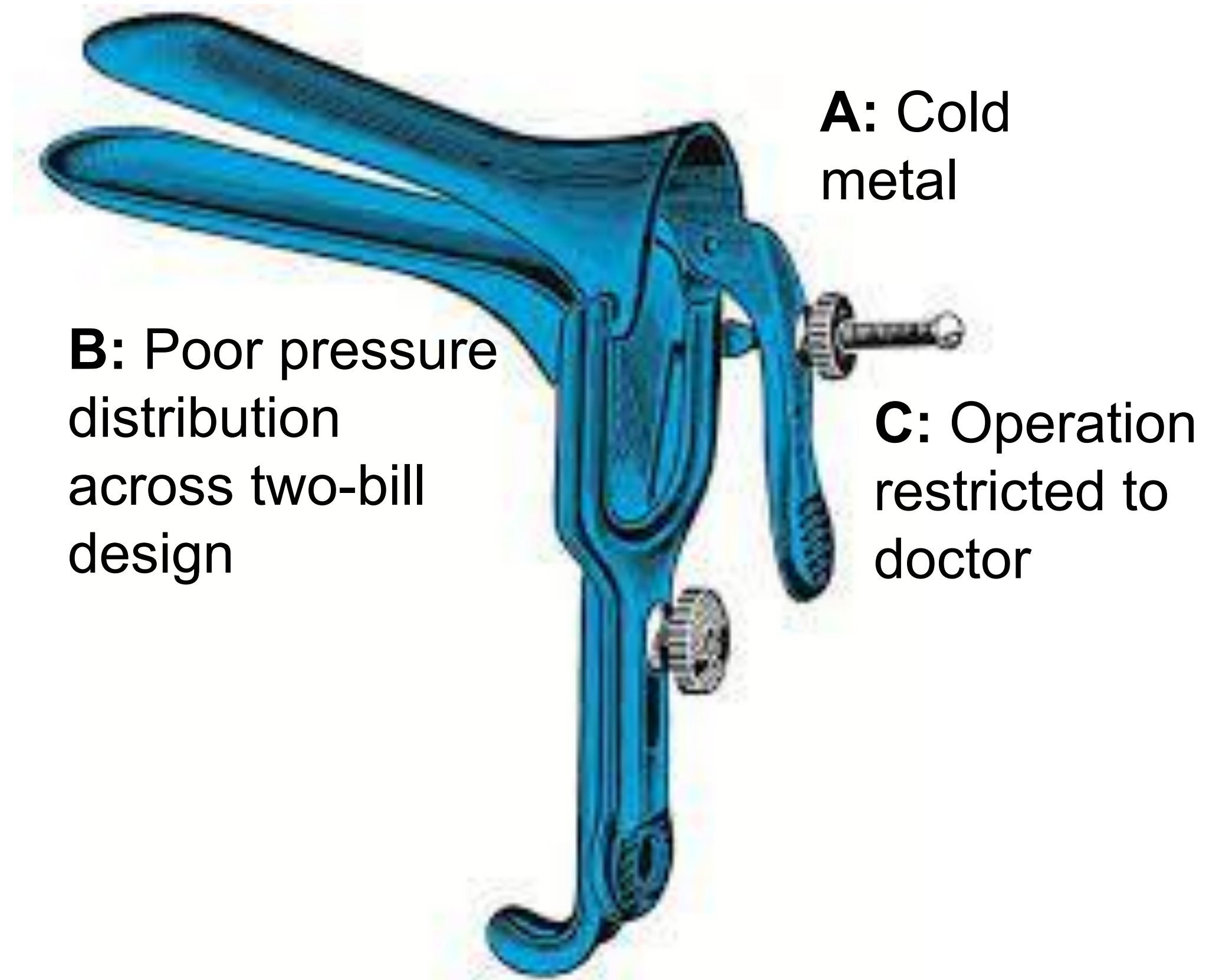
Pregnancy Ultrasound Wand



Patients are given the option to **insert the instrument themselves**

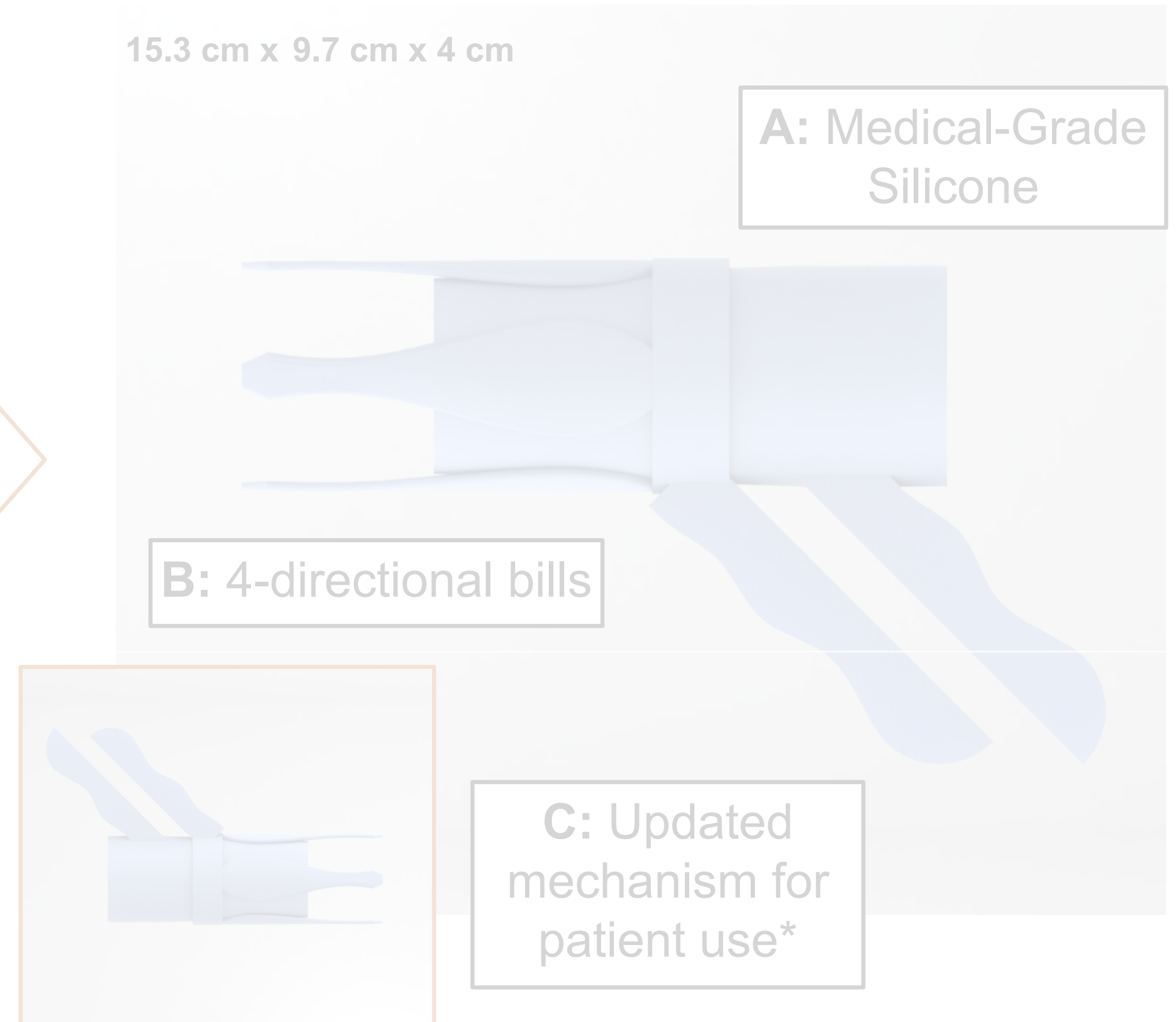
**** Clinician still present for examination**

Existing Design Flaws



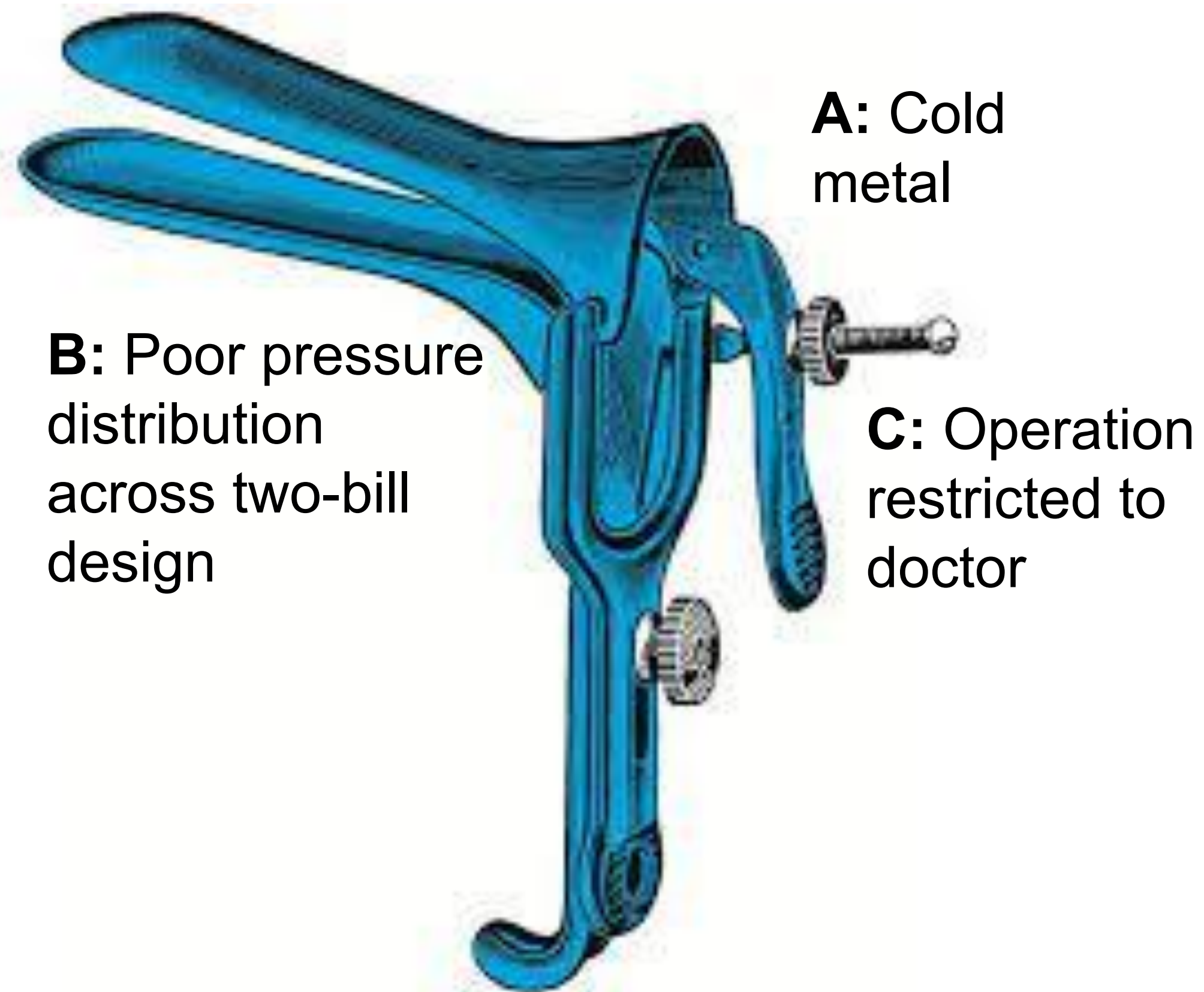
Solution

15.3 cm x 9.7 cm x 4 cm

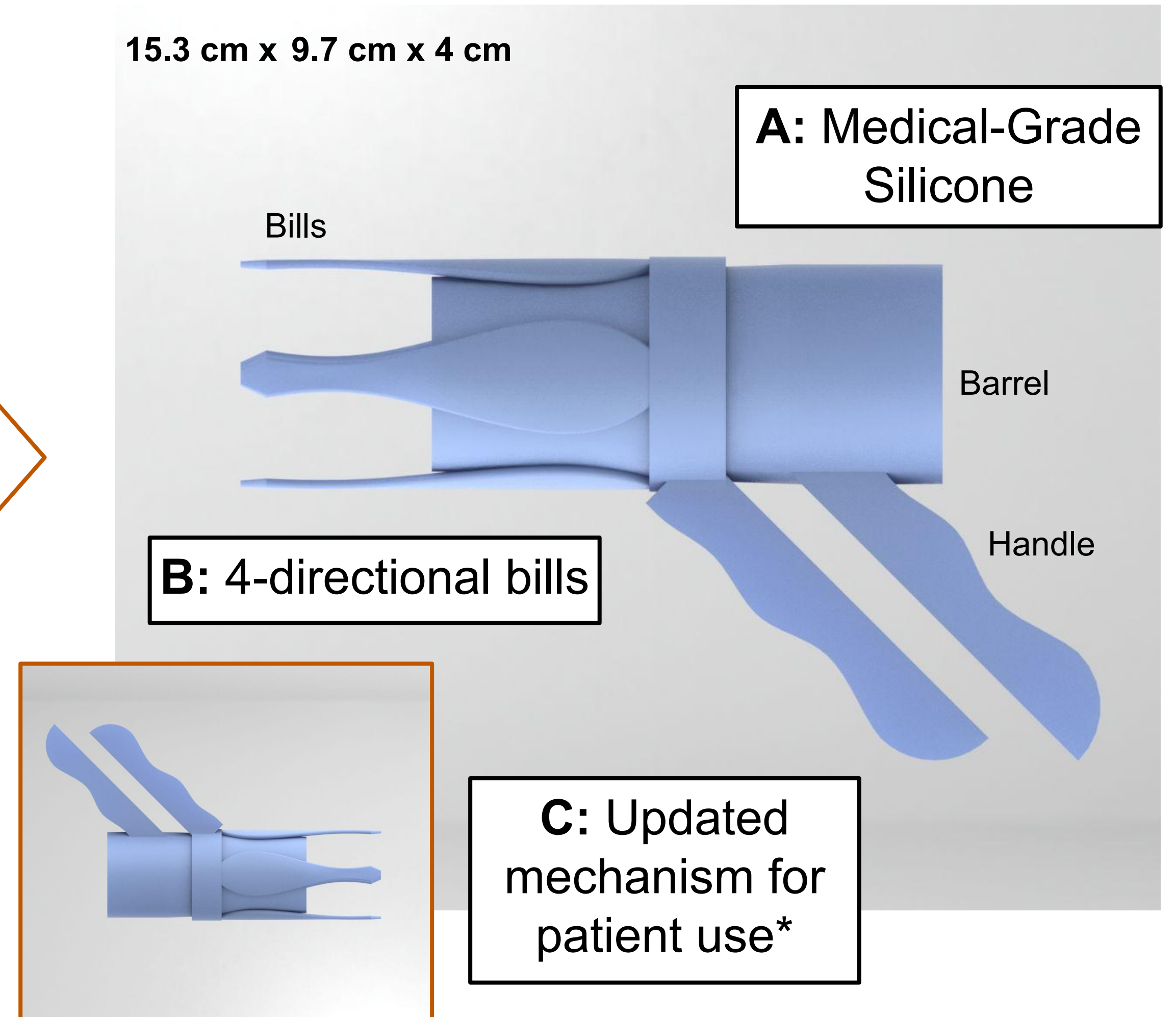


*Clinician still present

Existing Design Flaws

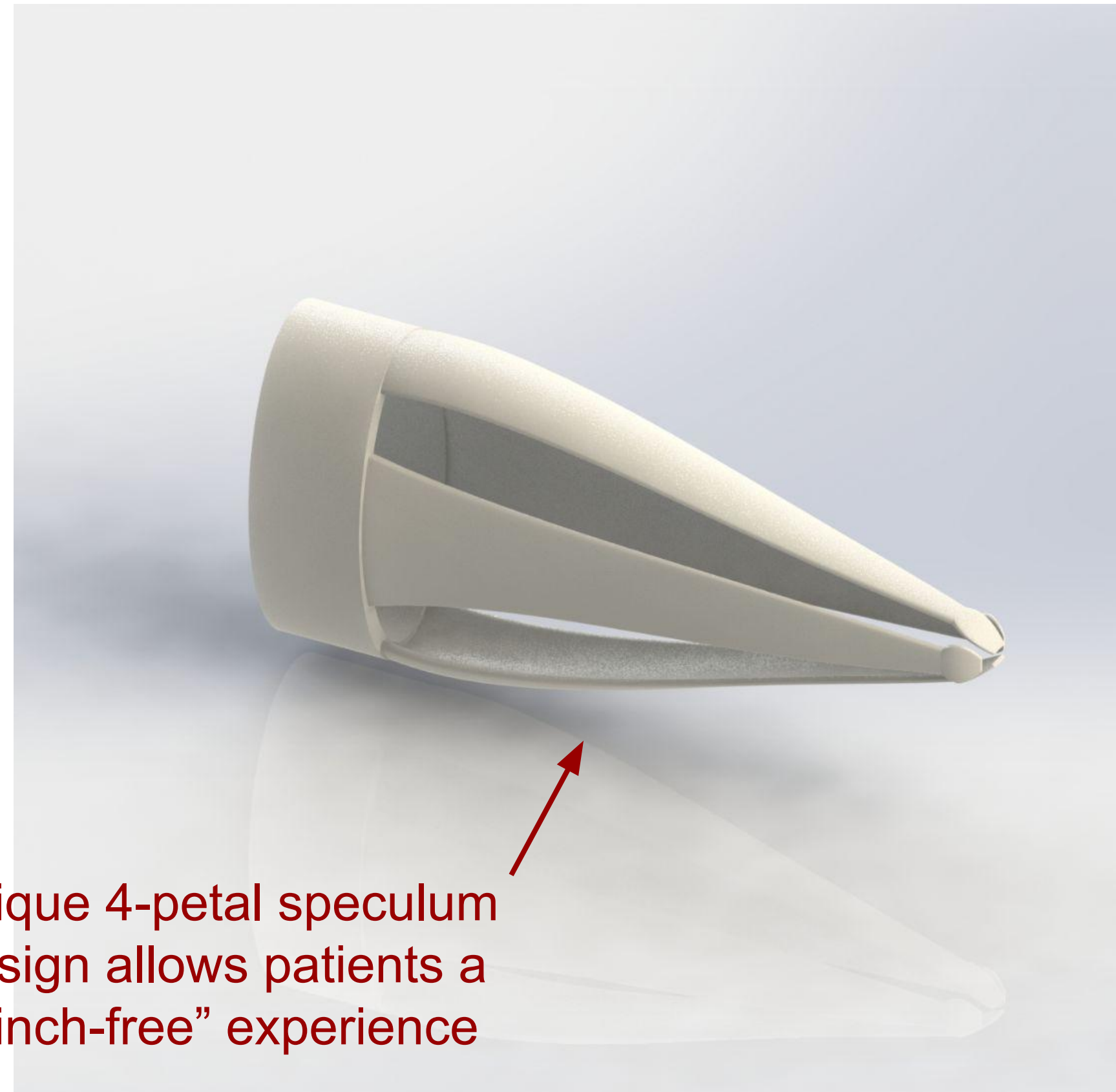


Solution

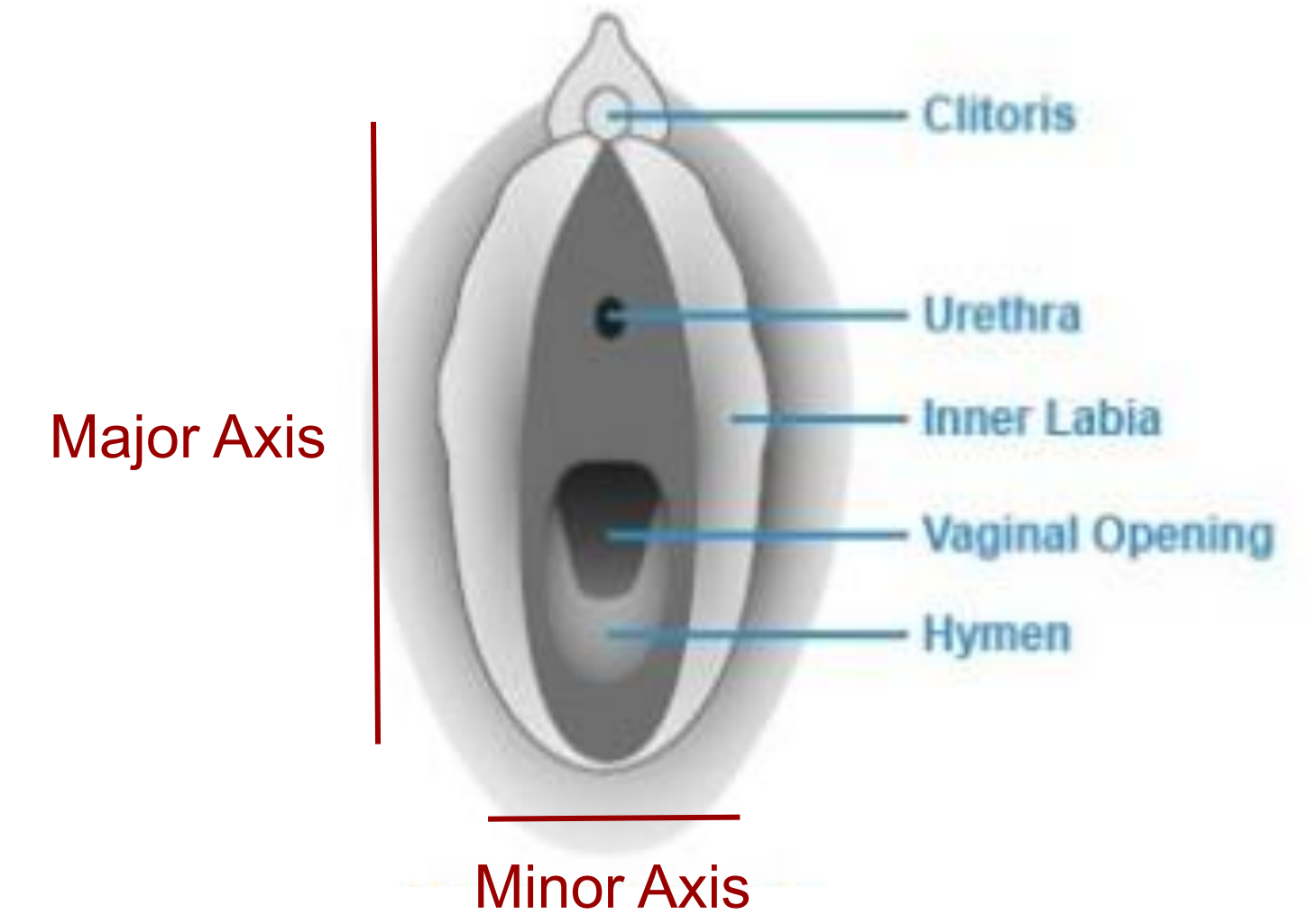
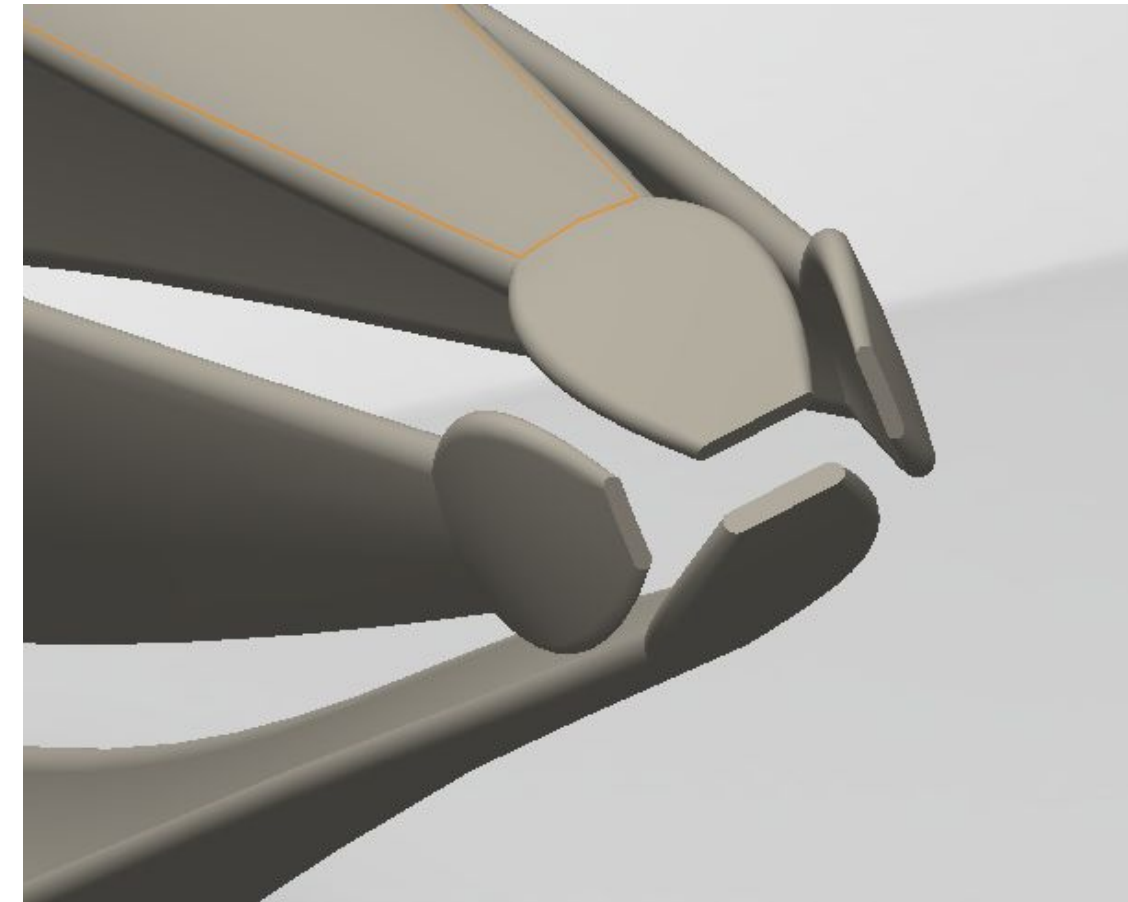


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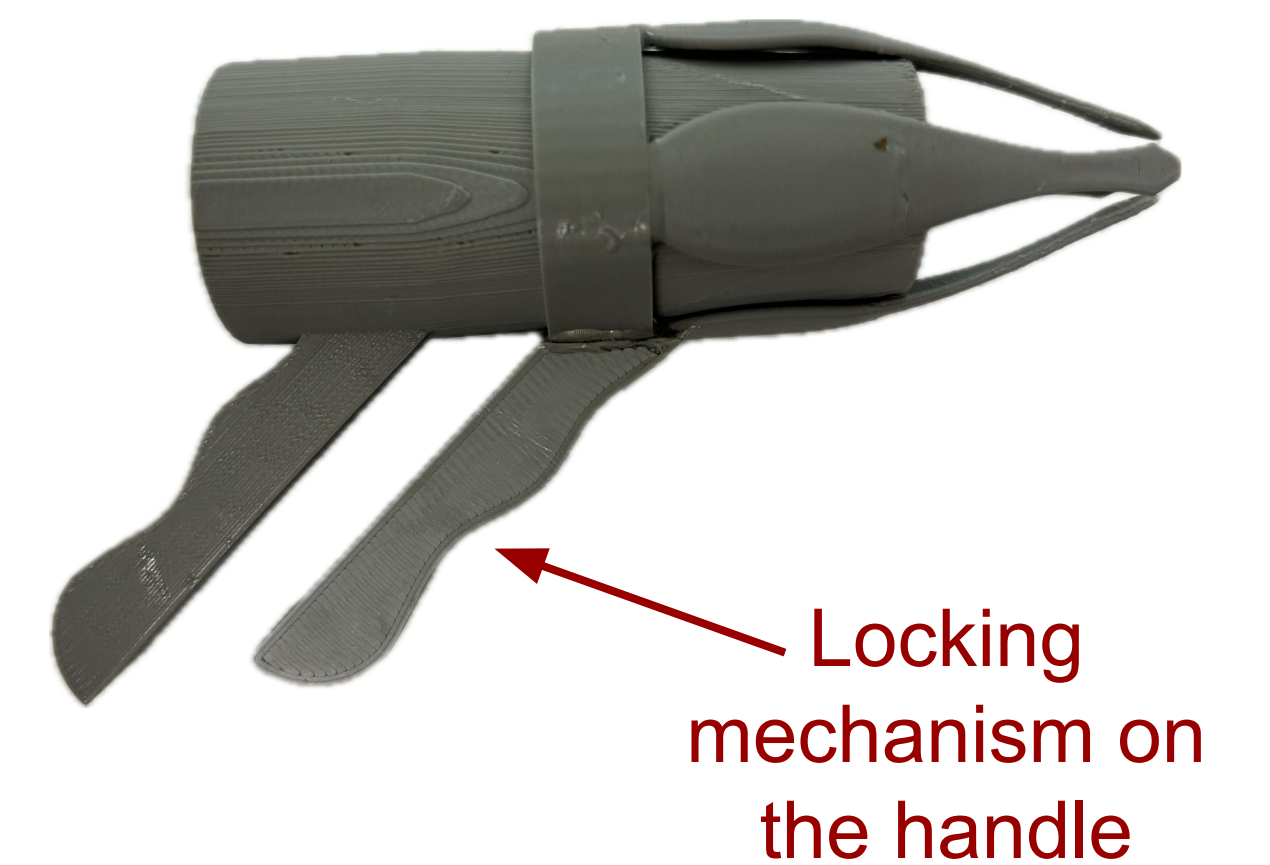
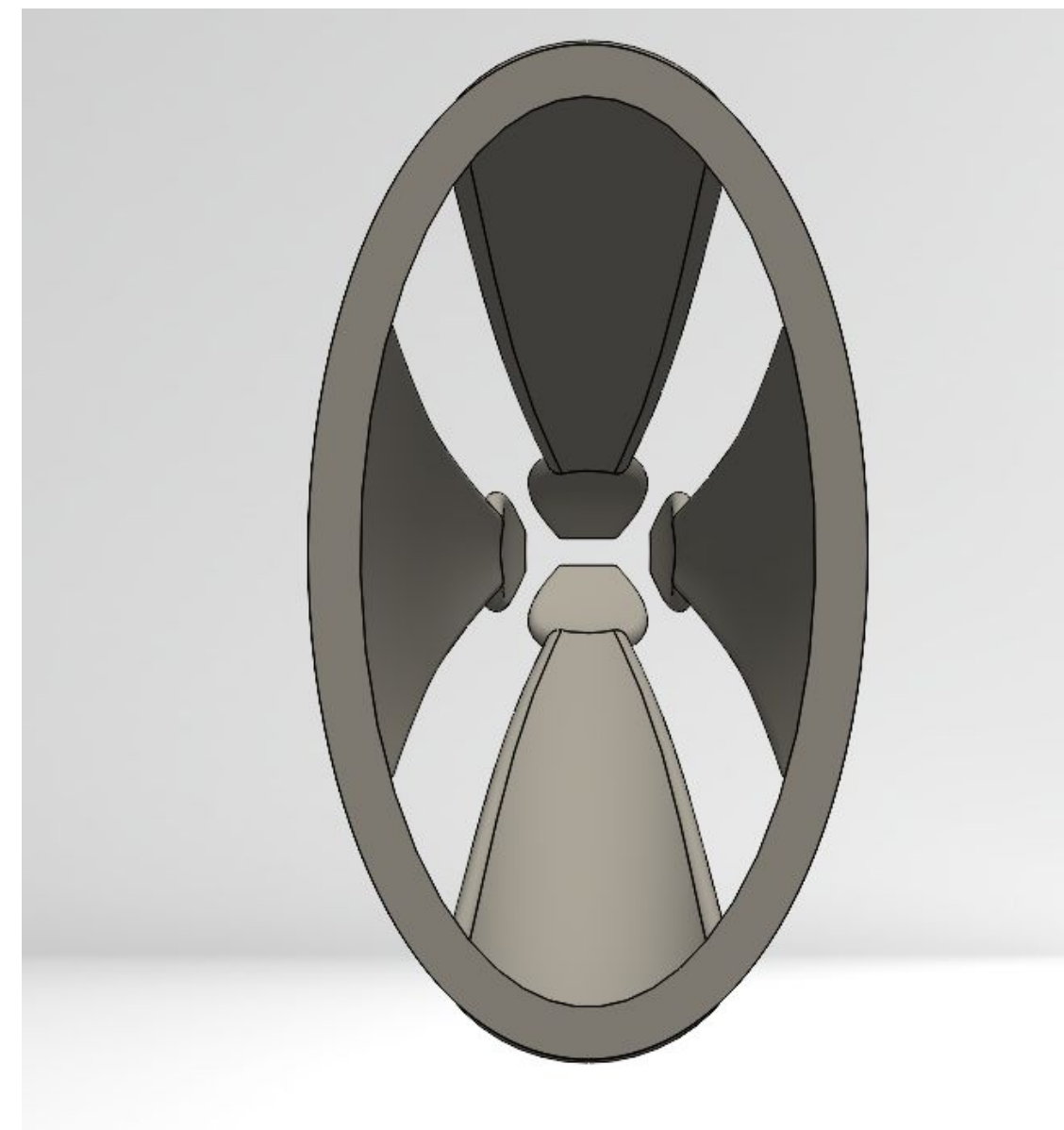
2nd Iteration of Bill Design



Small Profile Tip



Oval Shape



Materials:

Liquid Silicone Rubber

- 60-80 Shore A hardness
 - Stability upon insertion
- Medical Grade
 - Autoclave sterilizable
- Easy to store in bulk



Comparison to Metal

- Has higher cost per pound

However

- $\frac{1}{4}$ density compared to stainless steel 304
 - Lighter -> Comfort of use
- High tear/tensile strength
- Resists temperature change
- Increased material cost offset by reduced manufacturing costs

Manufacturing Process

Metal Speculum → \$\$\$

1. Forging
2. Machining
3. Polishing
4. Assembly
5. Sterilization



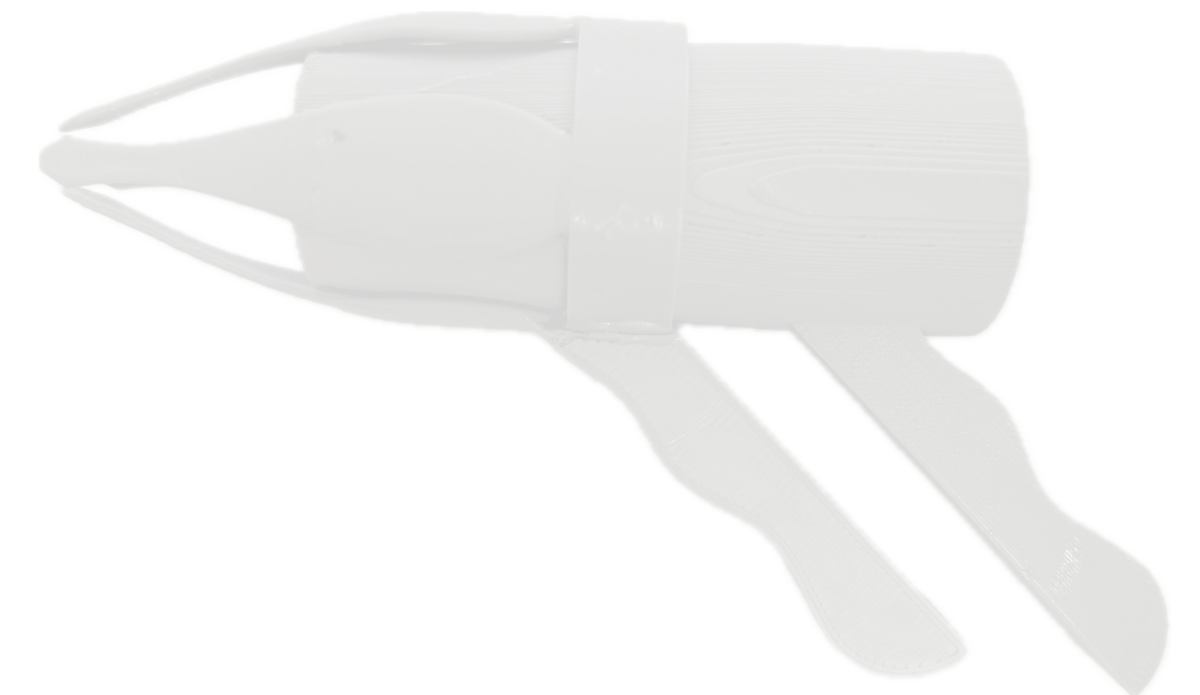
Plastic Speculum → \$\$

1. Injection molding
2. Assembly
3. Sterilization



Our Design: Silicon Speculum → \$

1. Injection molding
2. Sterilization



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1. Forging
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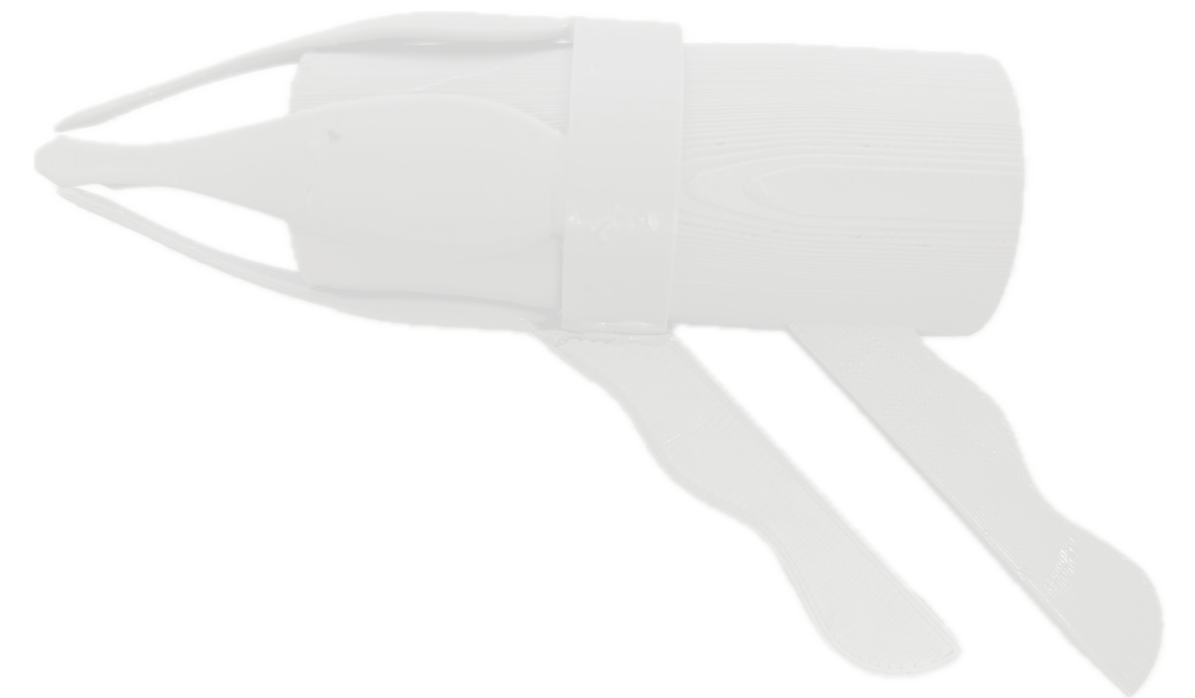
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FDA Approval

510(k) Route

- FDA has approved a **predicate device** with a multi-petal mechanism
- No clinical studies needed
 - mechanical testing will be performed
- **Silicon** is commonly used in other internal vaginal devices
 - menstrual cups and discs



Device Class

- FDA classifies **non-metal** vaginal speculum as **Class II**¹



1. <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpdc/classification.cfm>

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Improving User Experience

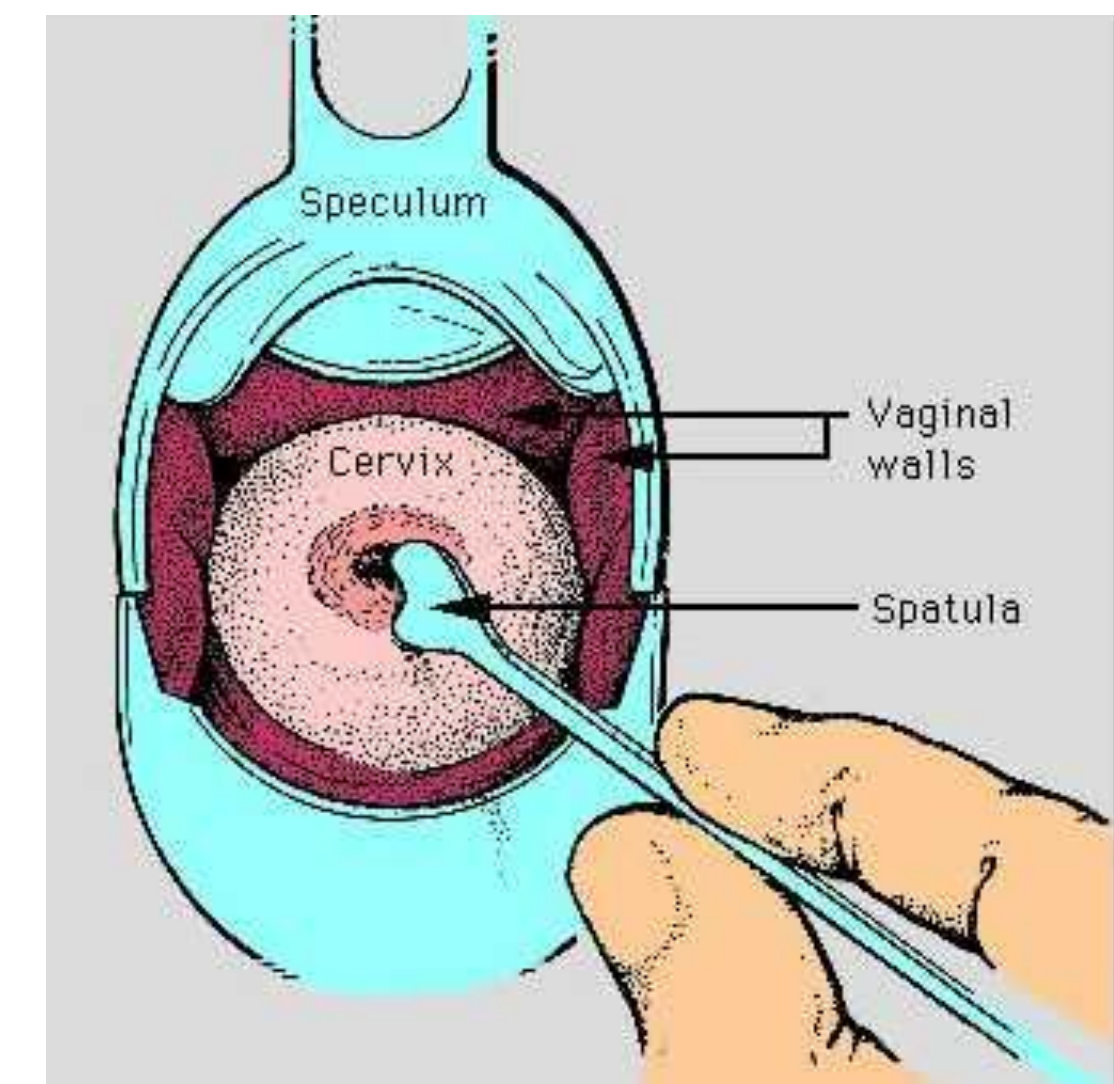
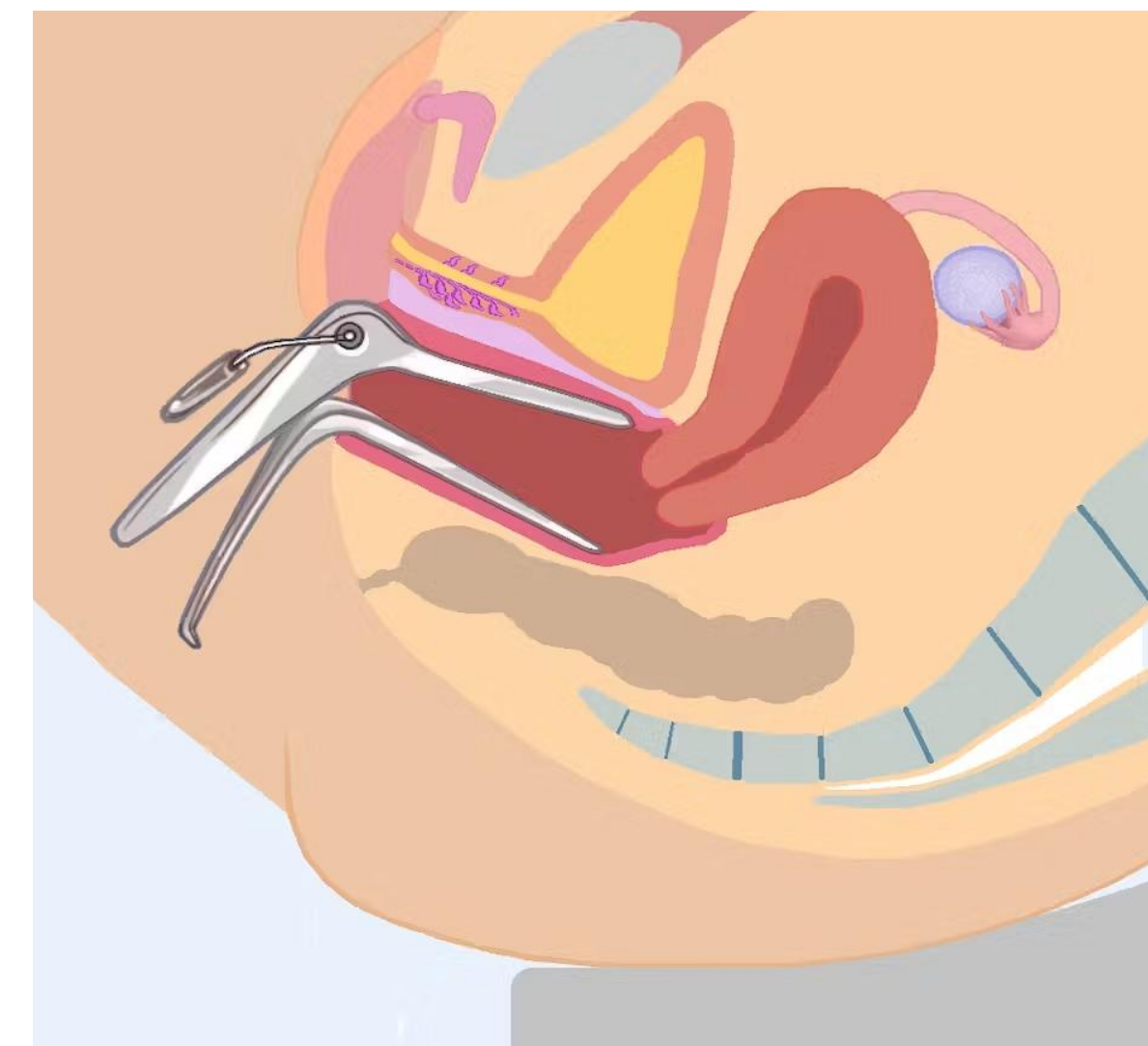
Need: anxiety and fear during the procedure

- ↪ Allow patient control over insertion of instrument
- ↪ Mitigate the invasive feeling of inserting the device



Need: physical discomfort and pain

- ↪ Uniform pressure on vaginal walls
- ↪ Gradual opening
- ↪ Medical-grade silicon



Improving User Experience

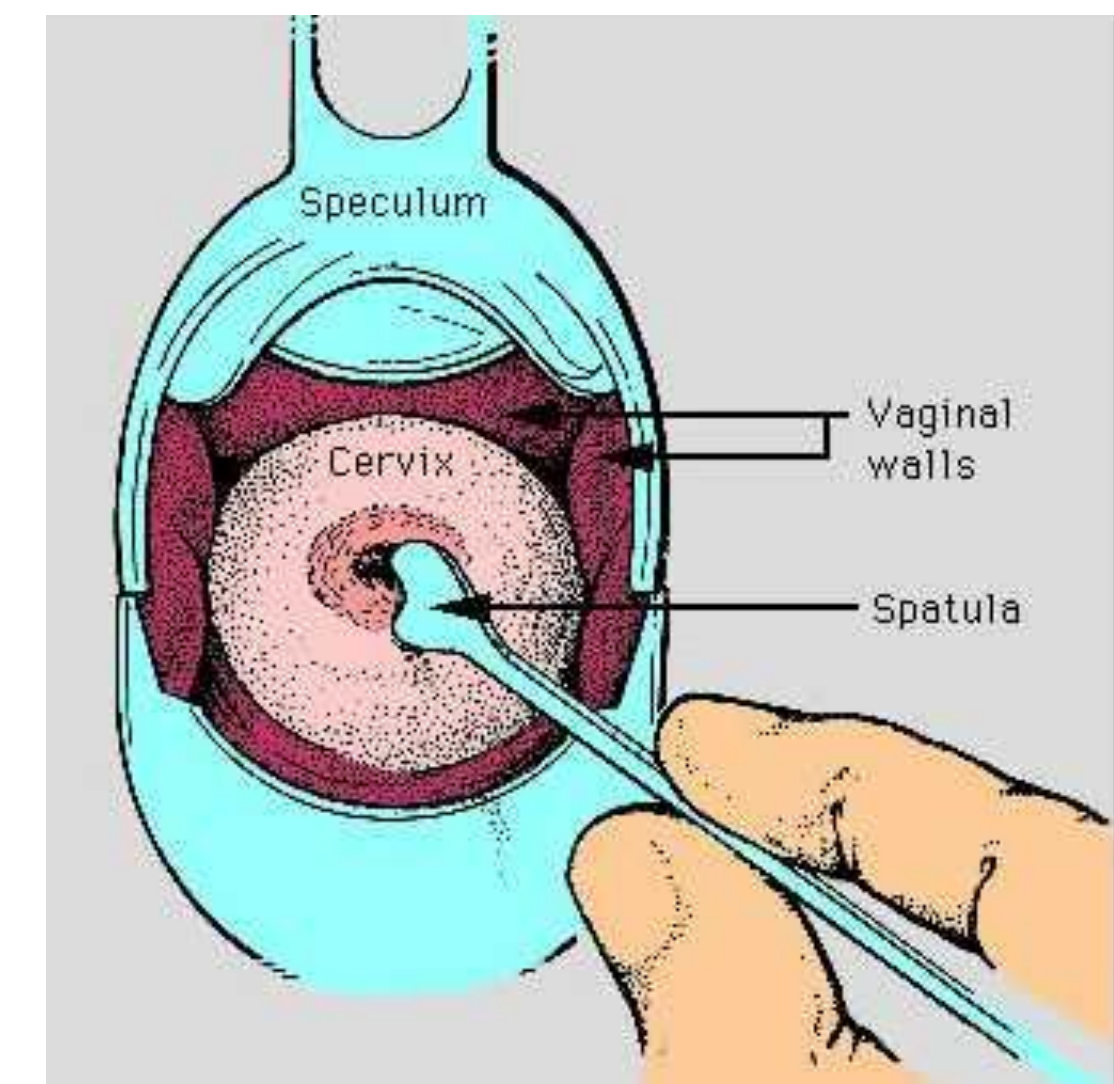
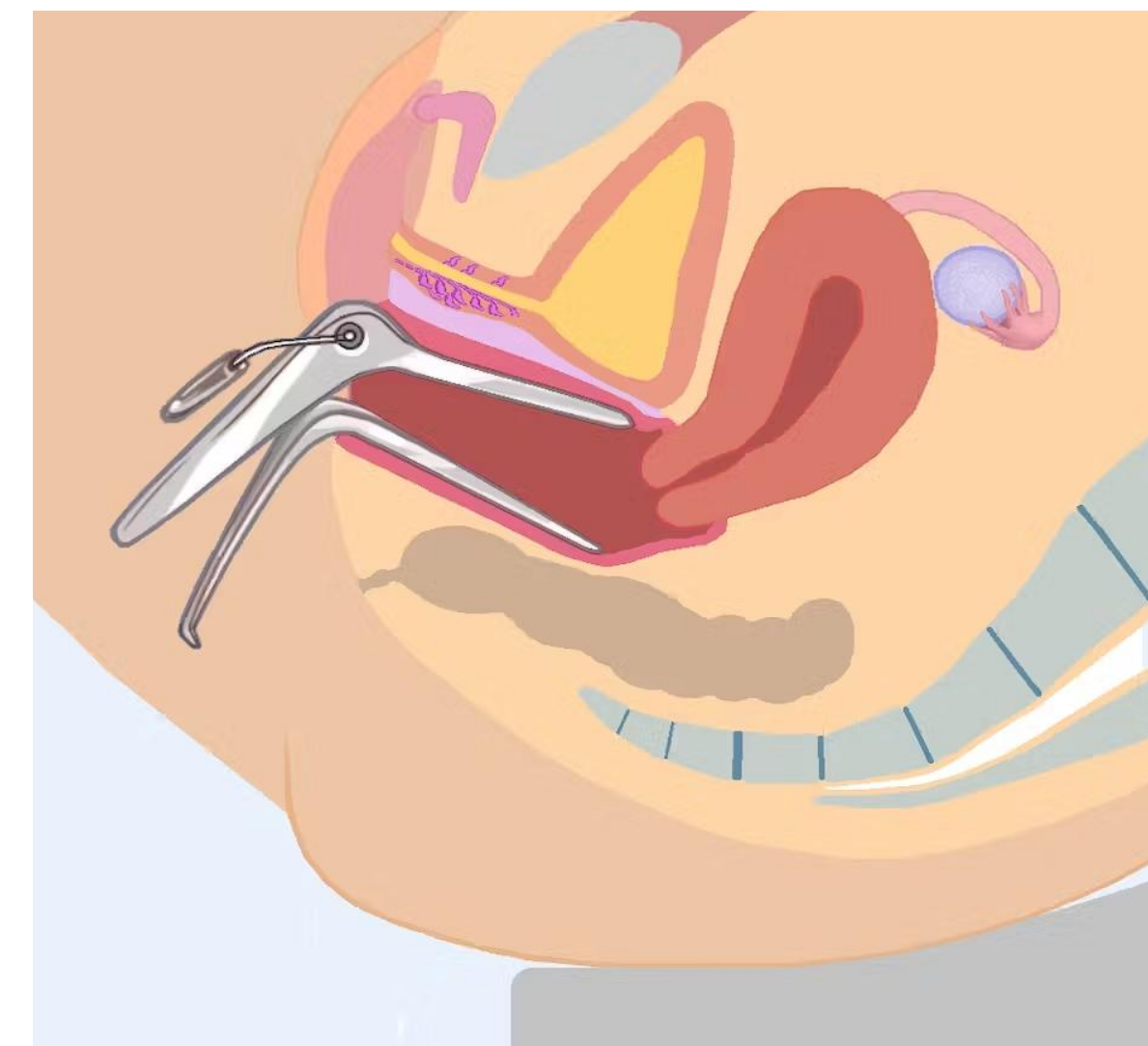
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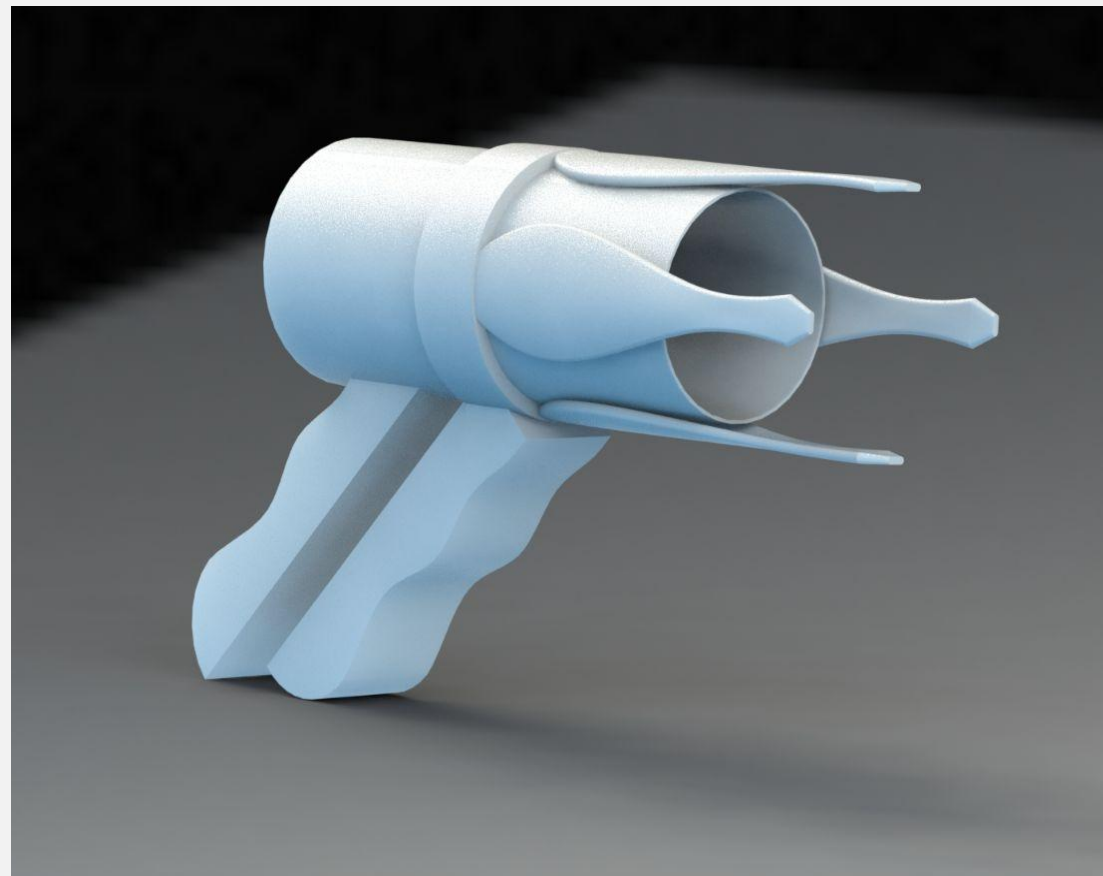


Patient **comfort** and **autonomy** are not at the center of the cervical examination process

Centering patient comfort will improve the experience for **all** patients and care providers during the procedure

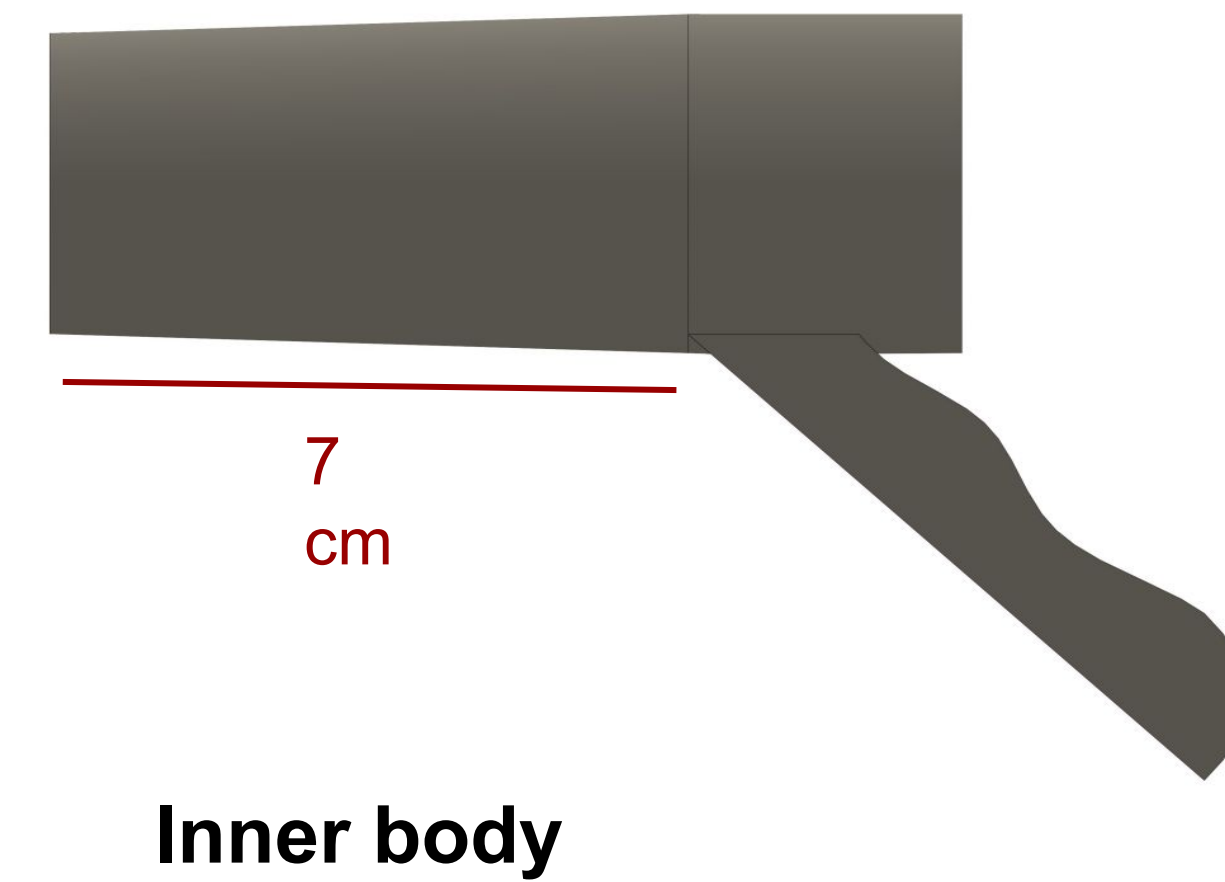
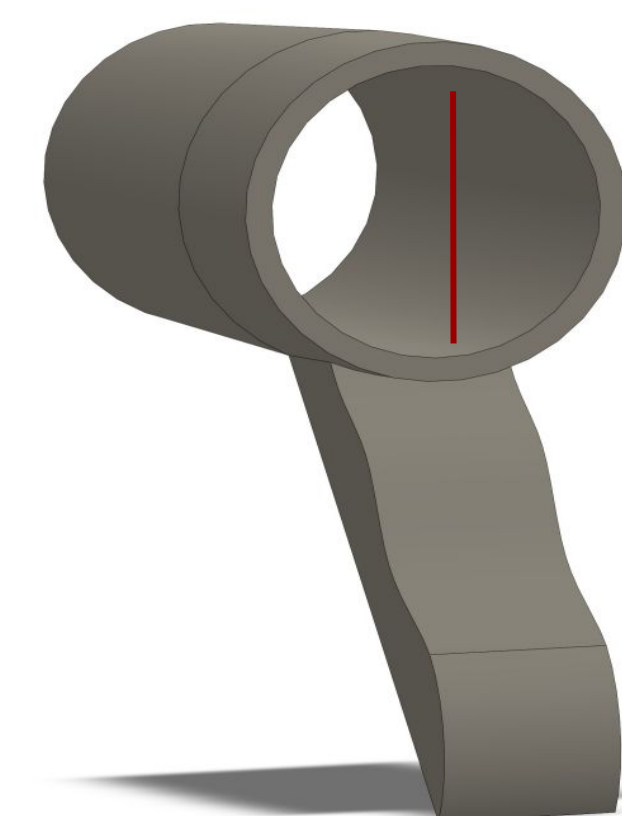
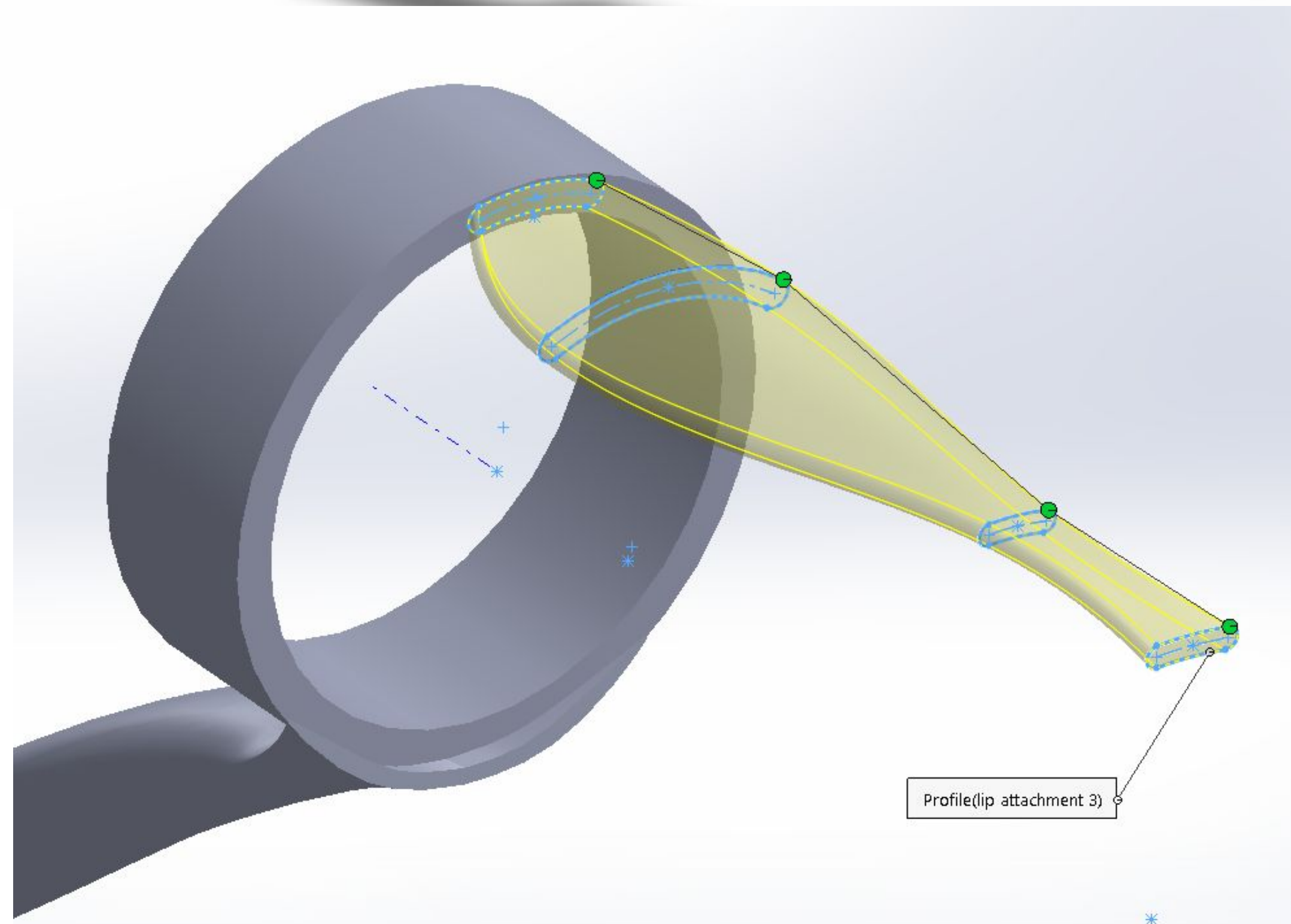
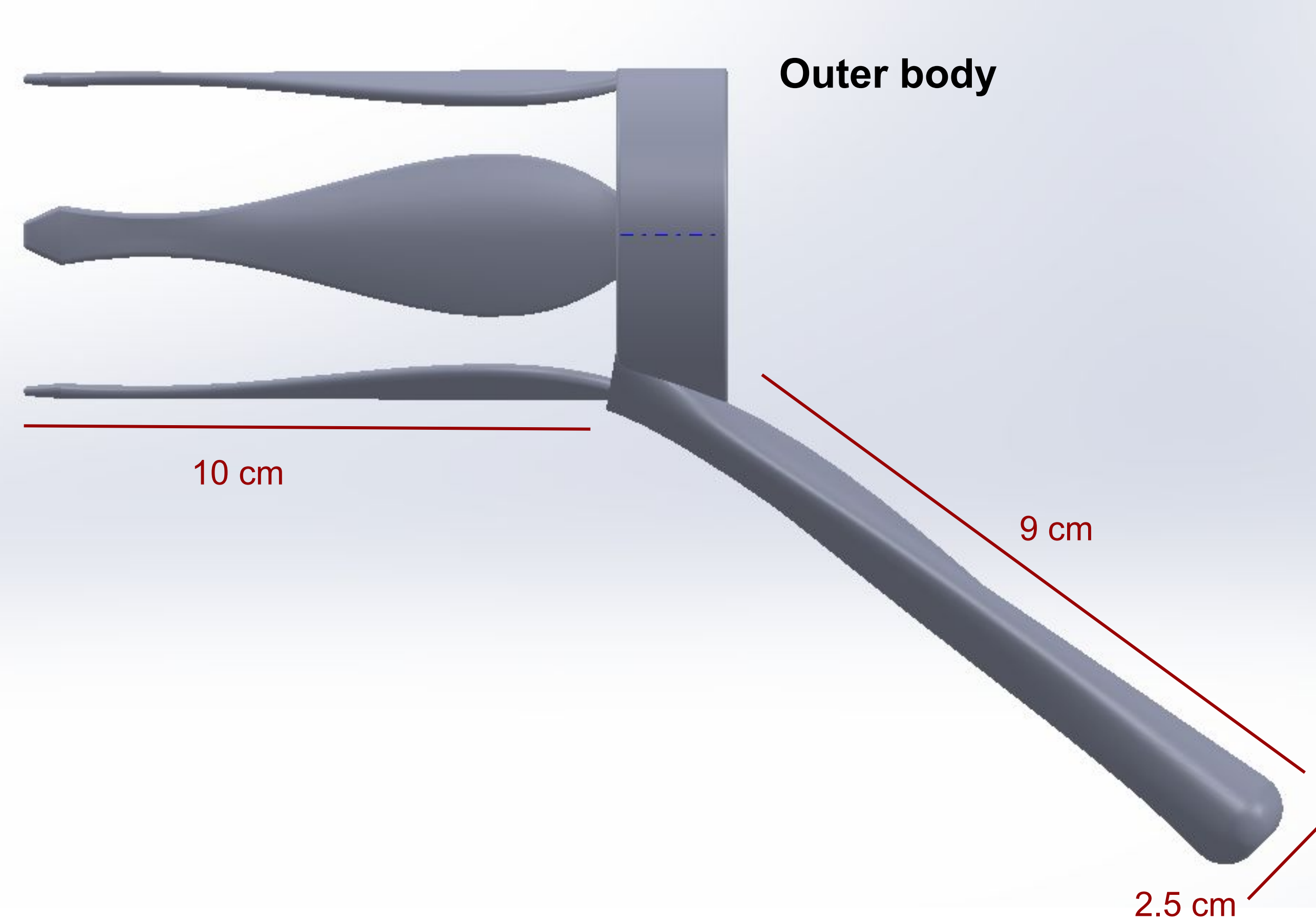
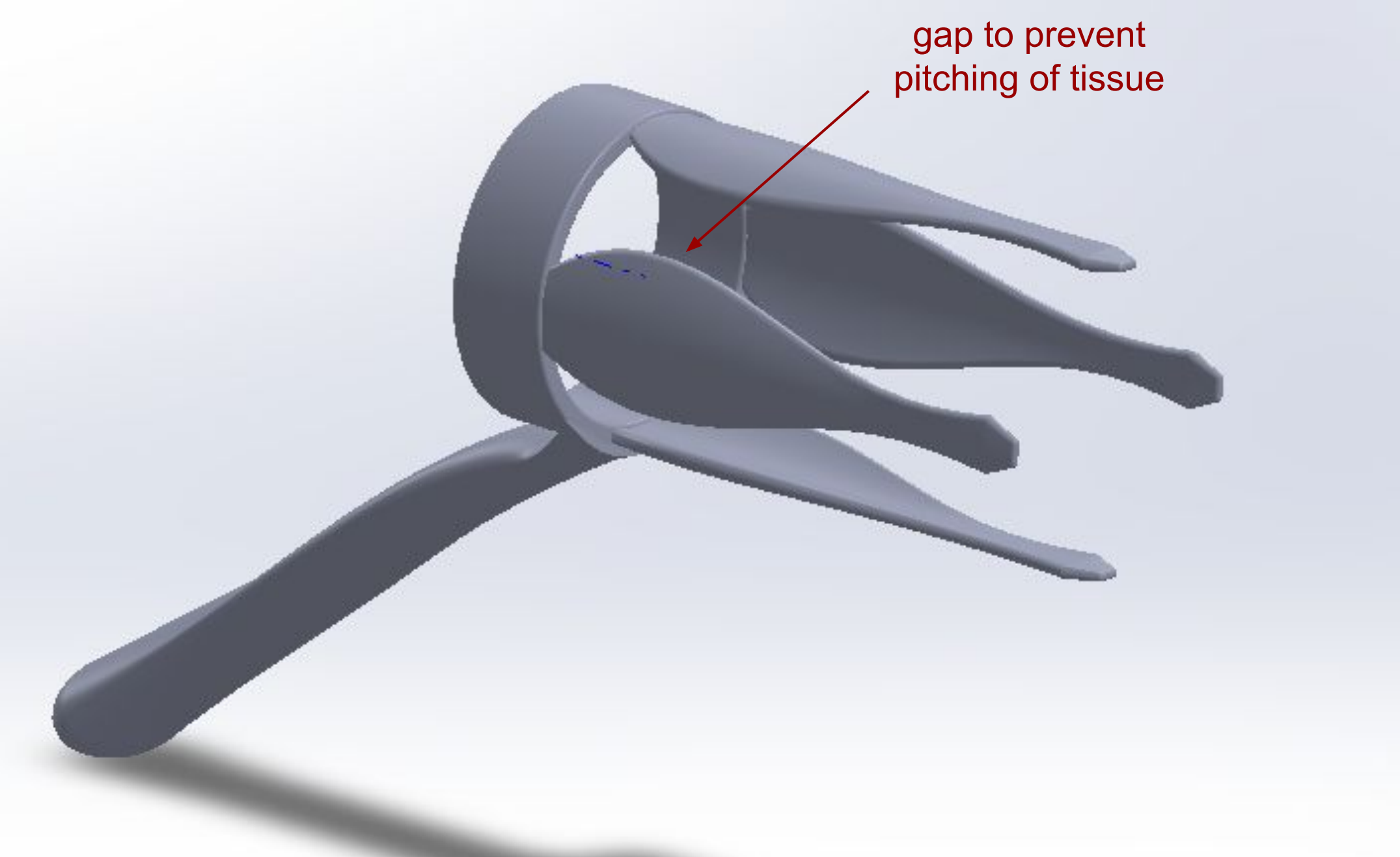
Questions?

Modified Speculum Design



3D Printed Design







Silicon in menstrual cups and discs



Speculum size references

- 1) <https://obgynmedicalantiques.com/vaginal-specula-1/graves-vaginal-speculum-c1878#:~:text=T.W.,in%20a%20base%20expanding%20instrument,https://us.edm-imaging.com/products/medgyn-disposable-led-vaginal-speculum>
- 2) <https://maternova.net/products/bouquet-speculum>
- 3) <https://my.clevelandclinic.org/health/diagnostics/4993-transvaginal-ultrasound>
- 4) <https://maternova.net/products/bouquet-speculum>

TEXAS ENGINEERING

Speculum: Market Analysis

Graves Speculum^{1,2}



Pros:

- Reusable
- Autoclavable
- FDA approved

Cons:

- Cold
- Uncomfortable
- Design has not changed significantly
- No patient control

Disposable Vaginal Speculum³



Pros:

- Clear
- Not cold

Cons:

- Uncomfortable
- Design has not changed significantly
- No patient control

Bouquet Speculum⁴



Pros:

- Newer more patient centric design
- Clear
- FDA approved

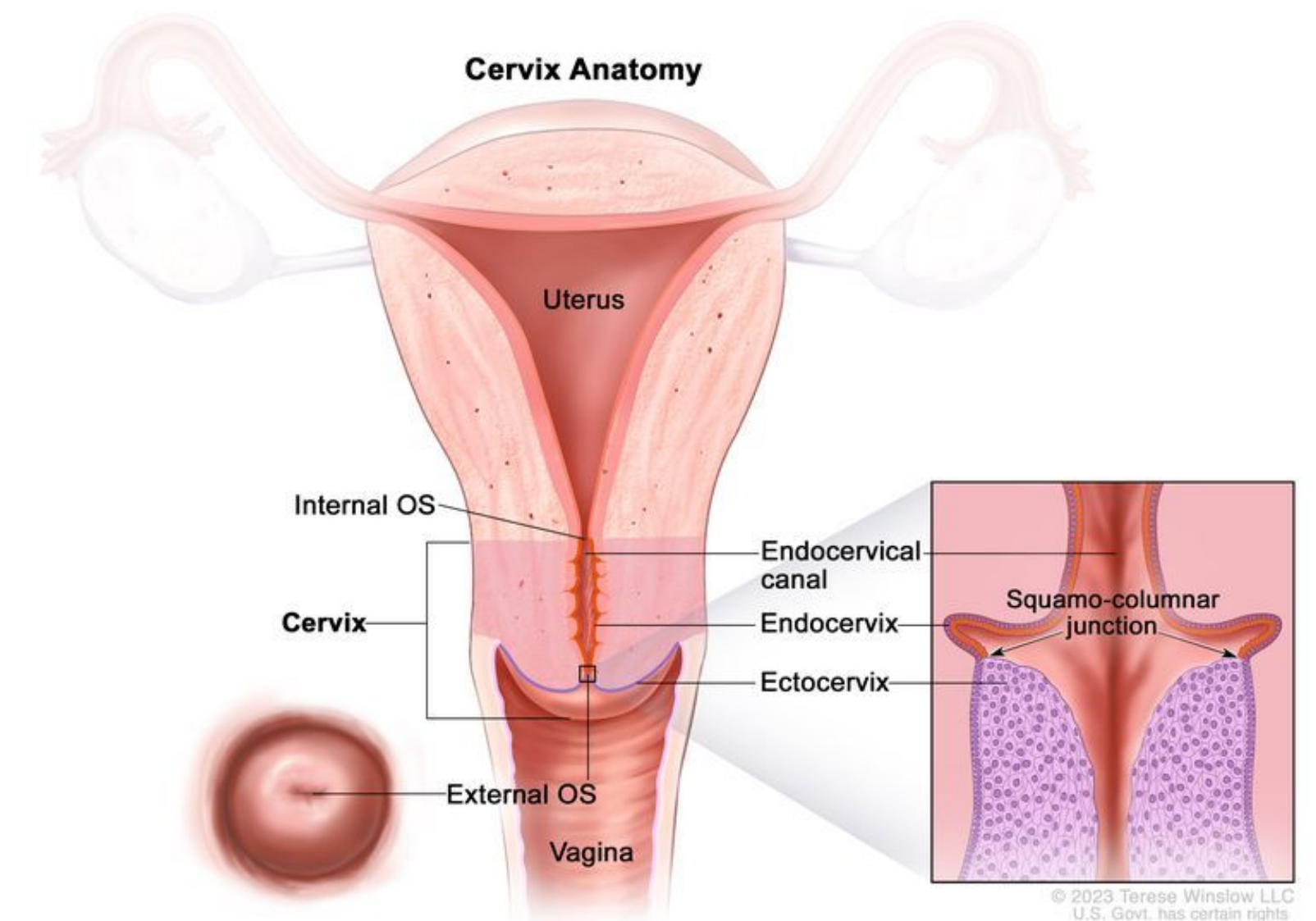
Cons:

- Not a established device
- Disposable
- No patient control

Background: Cervical Cancer

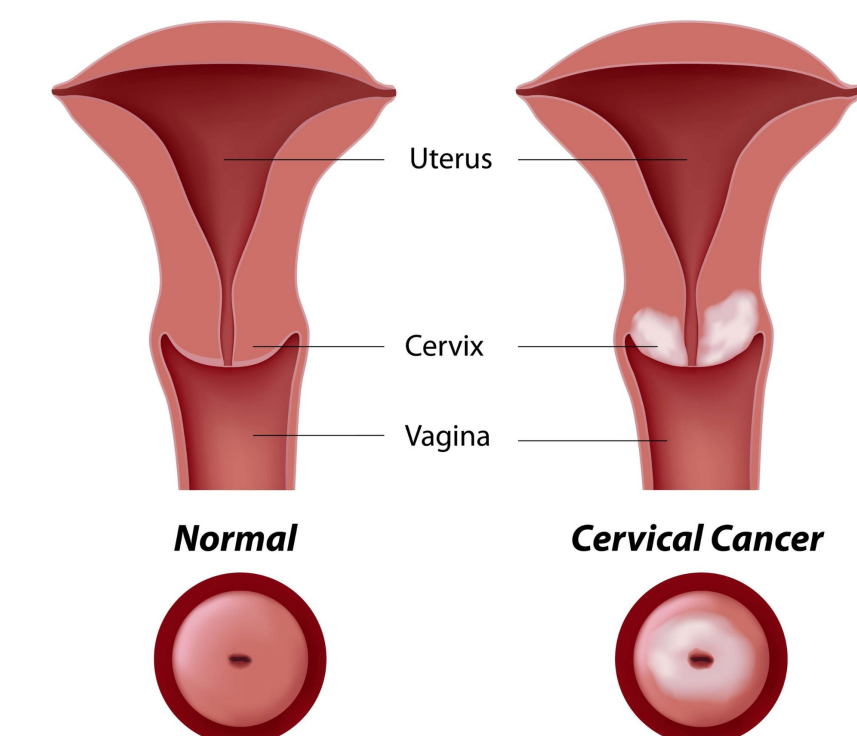
- Cervical cancer is the **fourth most common** malignancy among women worldwide¹
 - **569,000 new cases** each year²
- GLOBOCAN (WHO) estimated **342,000 deaths** from cervical cancer worldwide in 2020¹
- Cervical cancer has a long preclinical phase of decades without symptomatic effects³

Reproductive Anatomy



<https://www.cancer.gov/types/cervical>

Cervical Cancer



<https://www.hyperthermia-centre-hannover.com/cervical-cancer>